

Rockchip Developer Guide Linux GMAC Mode Configuration

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前言

概述

本文提供 Rockchip 平台以太网 GMAC 接口不同模式下的配置用例，用于解决以太网配置问题。

产品版本

芯片名称	内核版本
ROCKCHIP 芯片	所有版本

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	吴达超	2021-01-26	初始版本
V1.1.0	吴达超	2021-12-28	支持RK3588
V1.1.1	吴达超	2022-11-28	修正错误
V1.2.0	吴达超	2022-11-29	支持RK3528
V1.3.0	吴达超	2023-01-16	支持RK3562
V1.4.0	吴达超	2024-04-26	支持RK3576, 配置参考 RK3588

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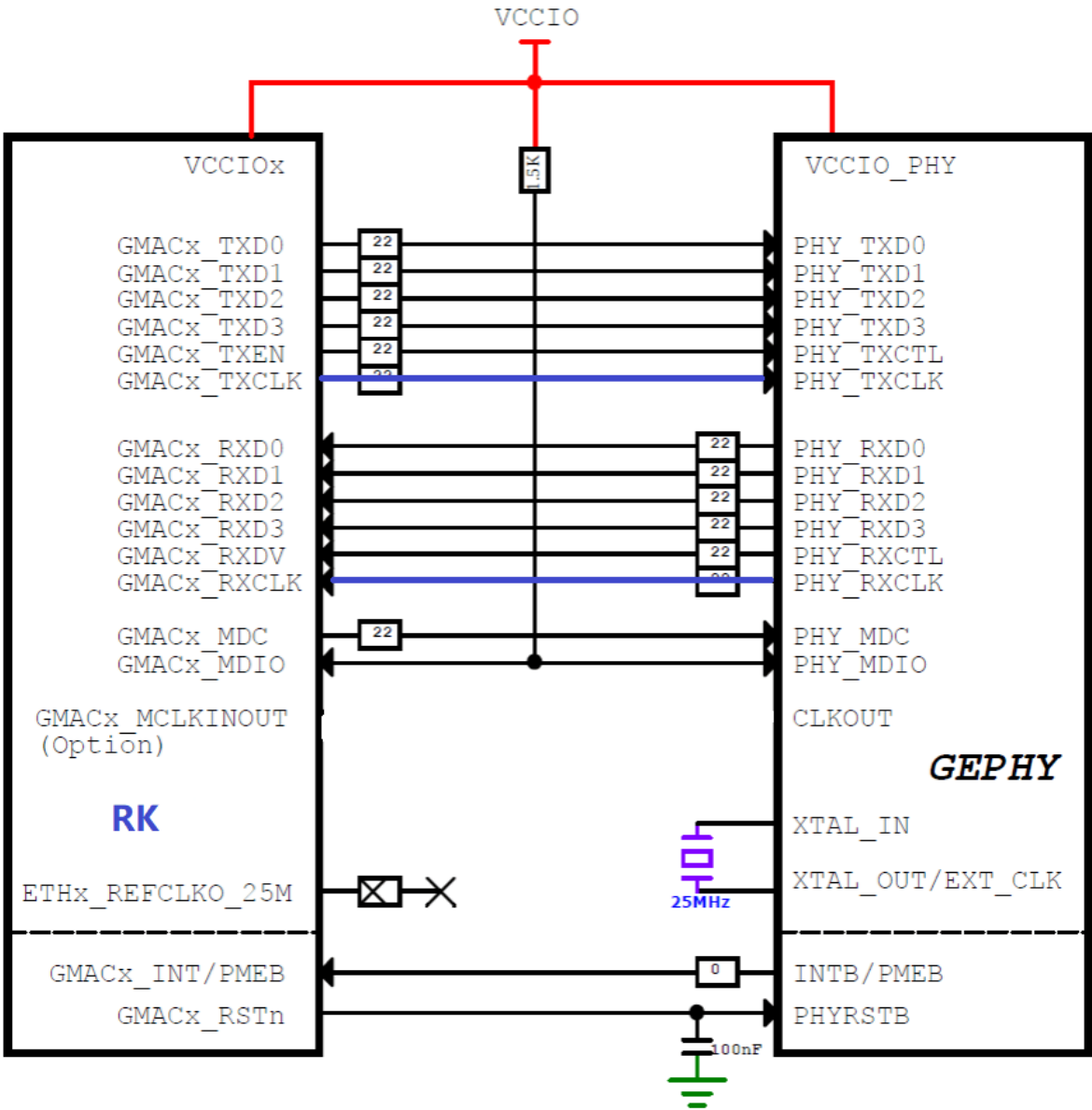
RGMII/RMII 主要模式连接图罗列如下：

1. RGMII 模式

一般使用主控 PLL 输出 时钟 output 方式，PHY 提供的 125M 时钟作为 input 方式为备选方案。

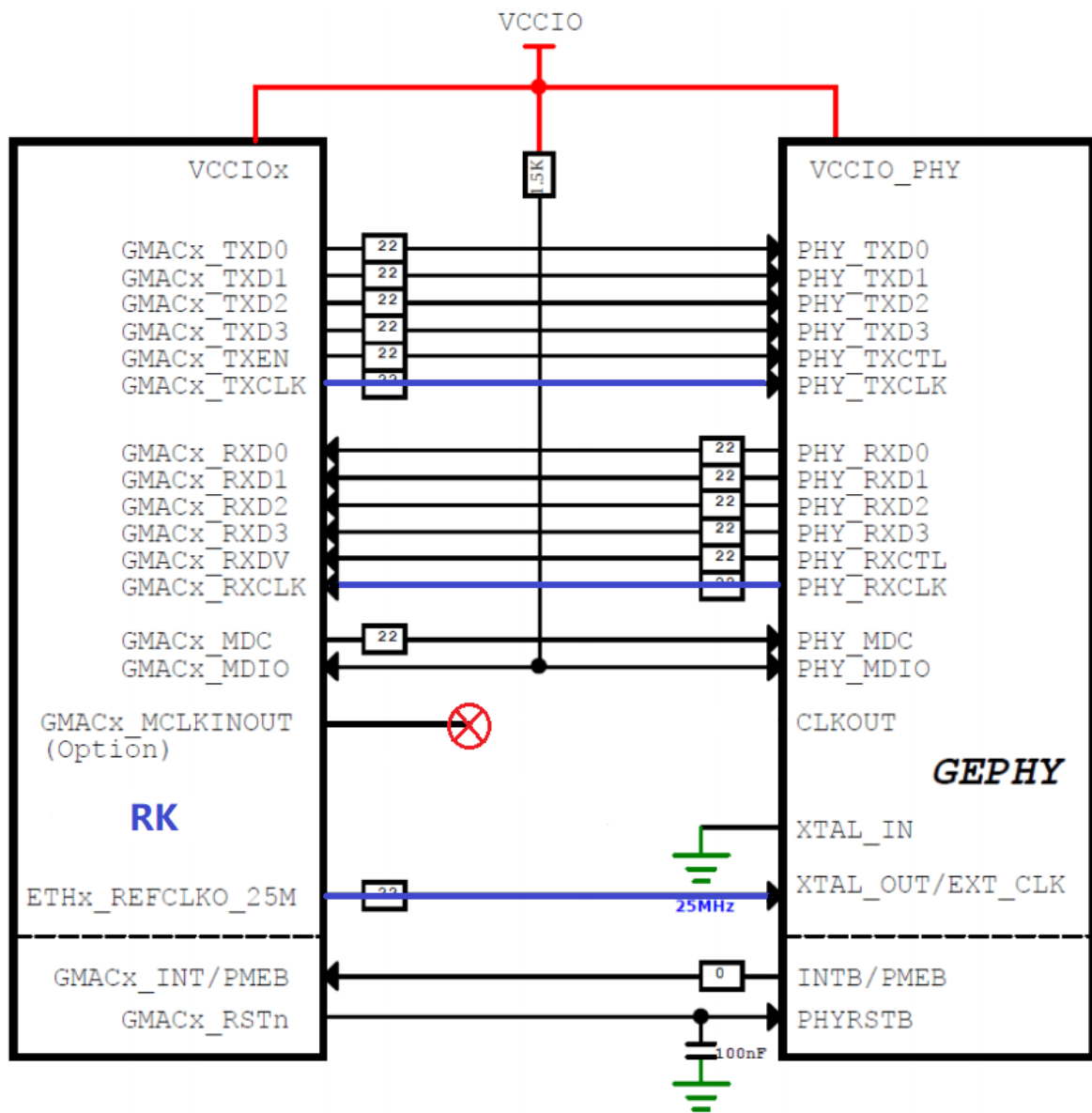
1.1 PLL output 125M for TX_CLK, Crystal 25M for PHY

主控 PLL 提供 TXCLK 所需时钟，PHY 25M 时钟由晶振提供。



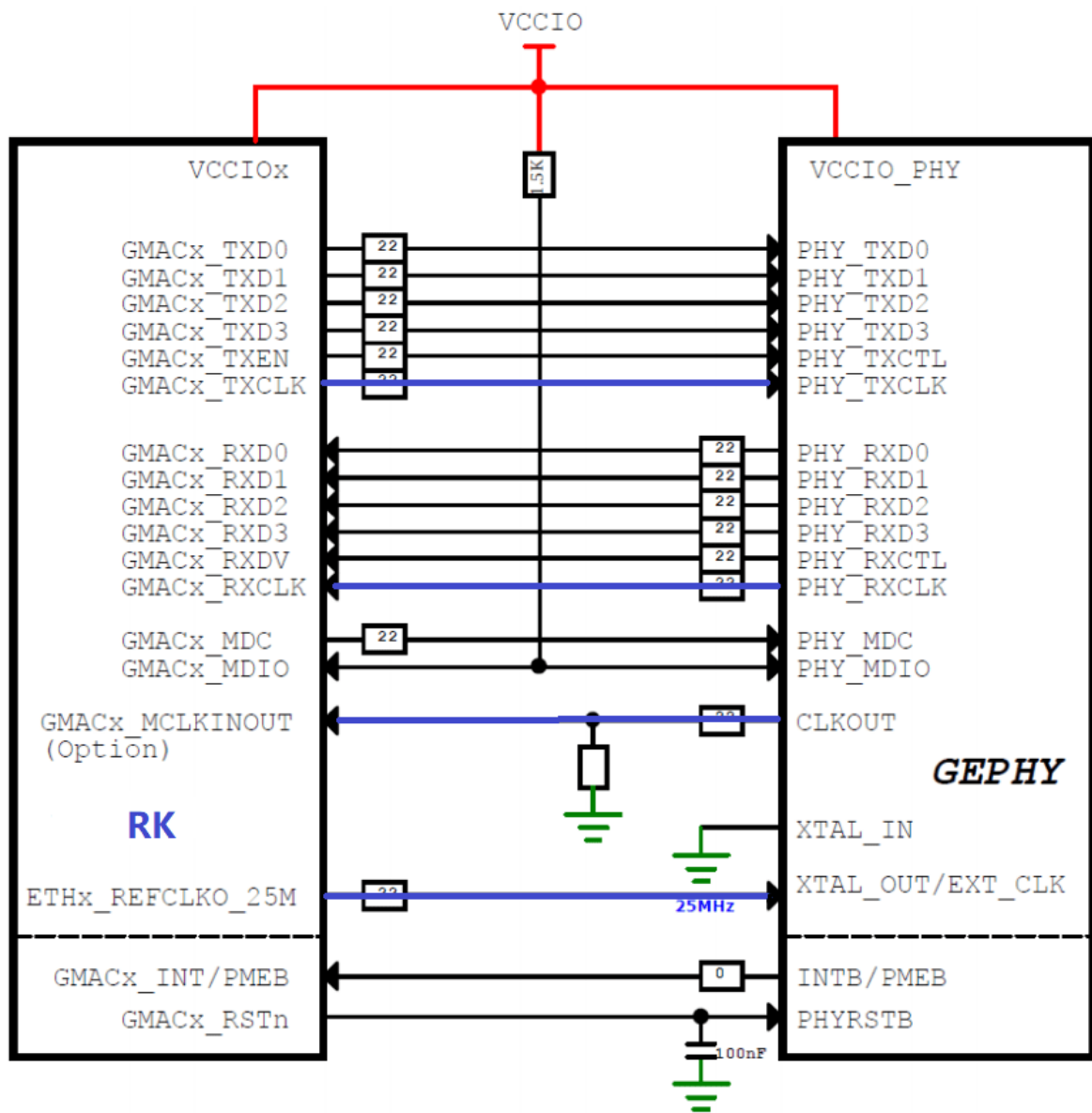
1.2 PLL output 125M for TX_CLK, PLL 25M for PHY

主控 PLL 提供 TXCLK 所需时钟，PHY 25M 时钟由主控提供。



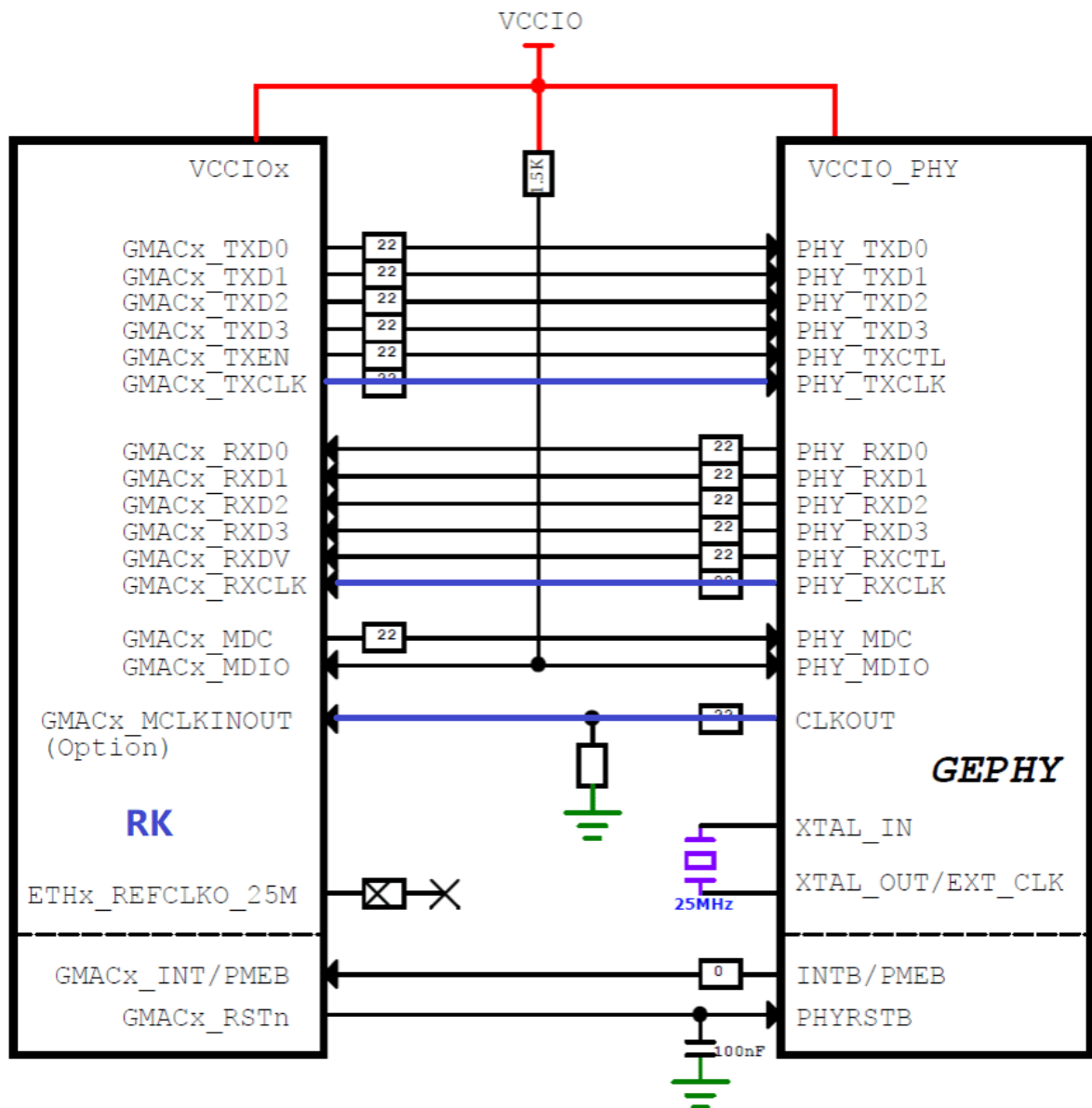
1.3 125M TX_CLK input from PHY, PLL 25M for PHY

TXCLK 所需时钟由 PHY 提供，PHY 25M 时钟由主控提供。



1.4 125M TX_CLK input from PHY, Crystal 25M for PHY

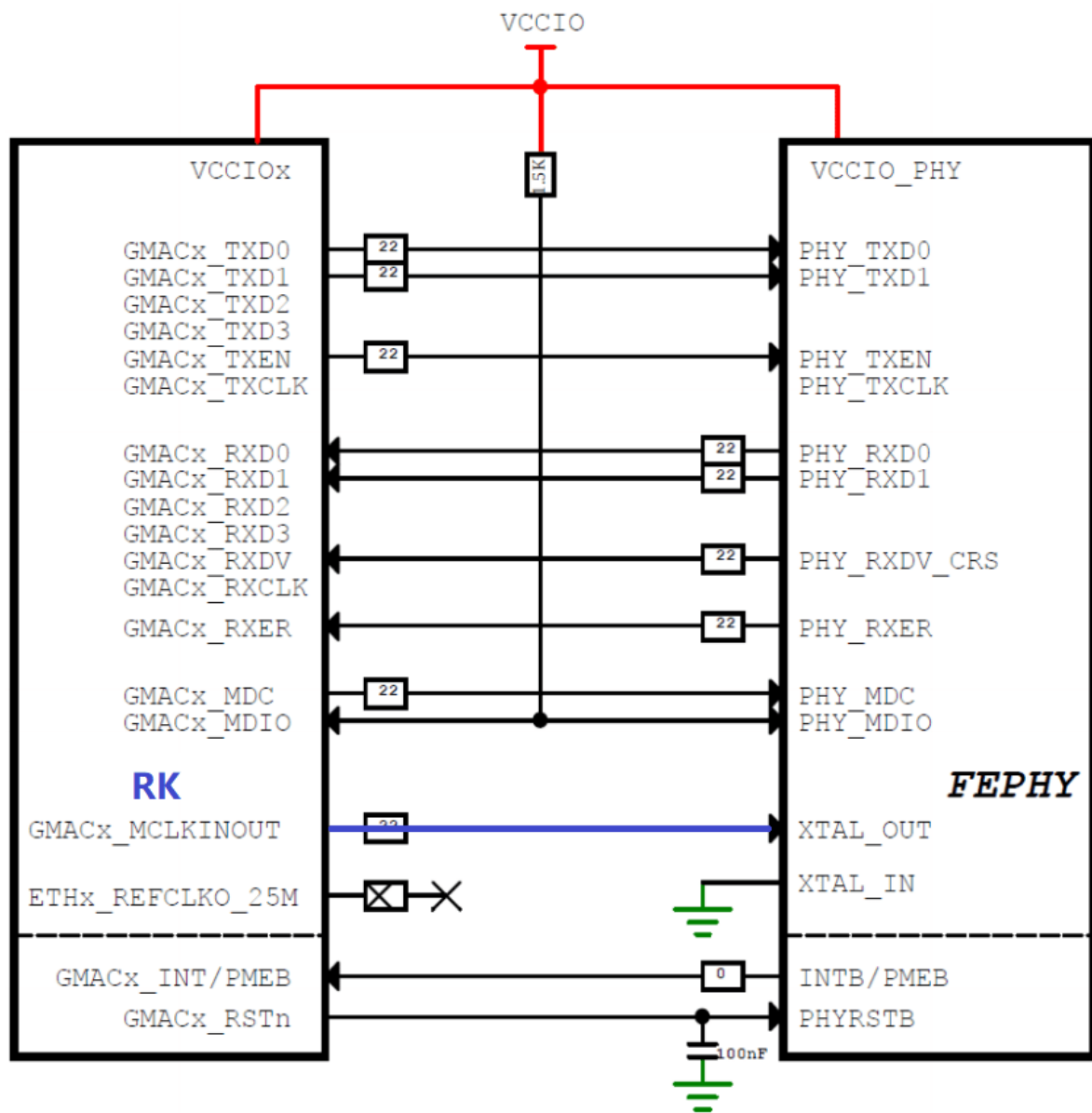
TXCLK 所需时钟由 PHY 提供，PHY 25M 时钟由晶振提供。



2. RMII 模式

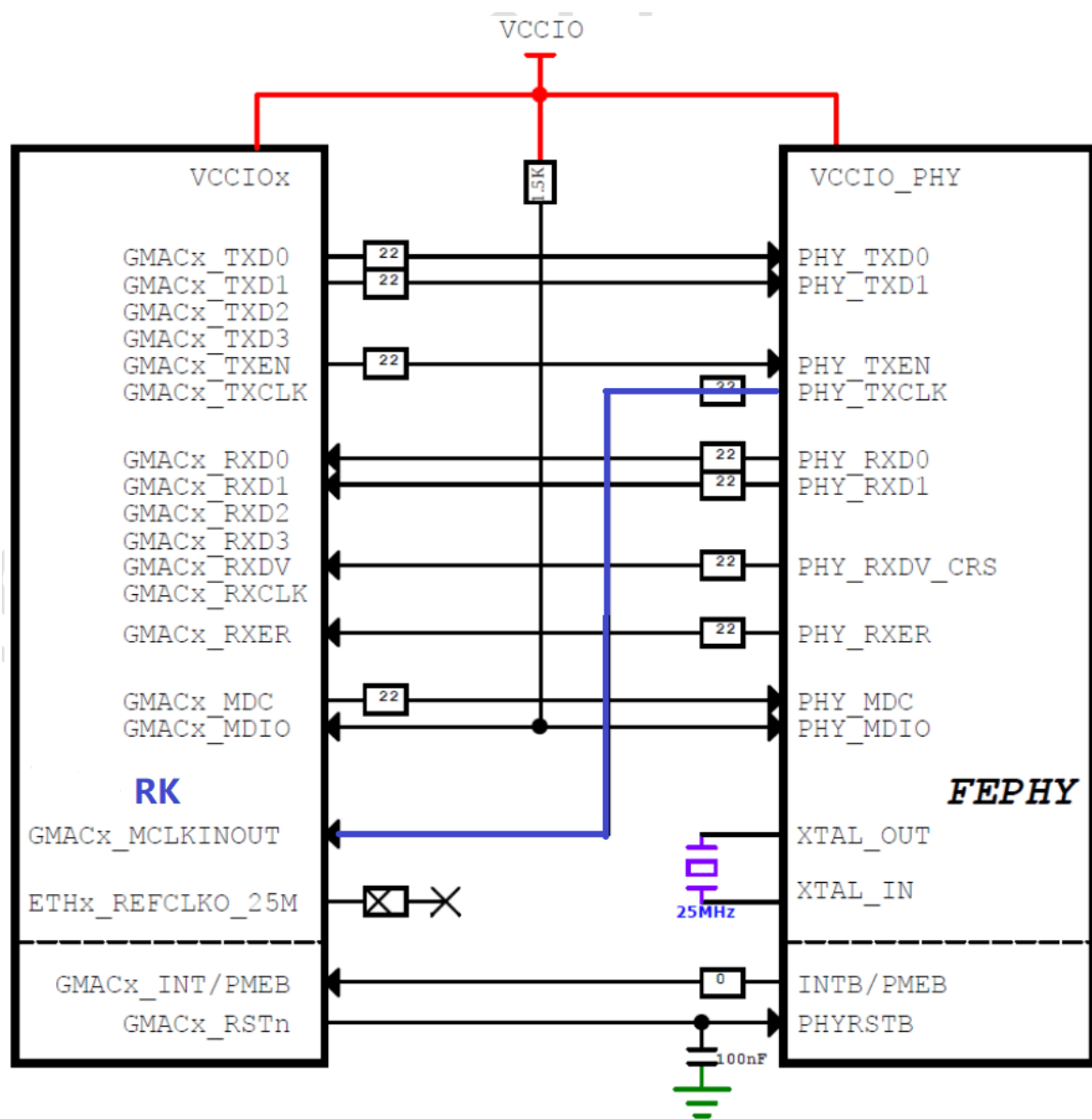
2.1 RMII Clock Output

主控提供 RMII 所需时钟



2.2 RMII Clock Input

PHY 提供 RMII 所需时钟



同样，RMII 模式下，晶振也可以由主控输出 25M 替代。

3. 模式配置

不同模式下的配置主要包含了 phy mode, clock 和 pinctrl 的配置，这些配置都是关联的，需要同时配置，否则无法工作。

以下是各芯片不同模式下，以 SDK 板级 DTS 为例的不同配置方式的参考，关注 dts 中 gmac 节点里 '+' 部分的修改。

3.1 PX30

3.1.1 RMII Clock Output

```

&gmac {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio2 13 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins &mac_refclk_12ma>;
    status = "okay";
};

```

3.1.2 RMII Clock Input

```

&gmac {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&gmac_clkin>;
    snps,reset-gpio = <&gpio2 13 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins &mac_refclk>;
    status = "okay";
};

```

3.2 RK1808

3.2.1 RMII Clock Output:

```

&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clocks = <&cru SCLK_GMAC>, <&cru SCLK_GMAC_RX_TX>,
+            <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC_REF>,
+            <&cru SCLK_GMAC_REFOUT>, <&cru ACLK_GMAC>,
+            <&cru PCLK_GMAC>, <&cru SCLK_GMAC_RMII_SPEED>;
+    clock-names = "stmmaceth", "mac_clk_rx",
+                 "mac_clk_tx", "clk_mac_ref",
+                 "clk_mac_refout", "aclk_mac",
+                 "pclk_mac", "clk_mac_speed";
+    assigned-clocks = <&cru SCLK_GMAC_RX_TX>;
+    assigned-clock-parents = <&cru SCLK_GMAC_RMII_SPEED>;
    snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};

```

```
};
```

3.2.2 RMII Clock Input

```
+&gmac_clkin {
+    clock-frequency = <50000000>;
+};

&gmac {
    phy-supply = <&vcc_phy>;
    phy-mode = "rmii";
+    clock_in_out = "input";
+    clocks = <&cru SCLK_GMAC>, <&cru SCLK_GMAC_RX_TX>,
+            <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC_REF>,
+            <&cru SCLK_GMAC_REFOUT>, <&cru ACLK_GMAC>,
+            <&cru PCLK_GMAC>, <&cru SCLK_GMAC_RMII_SPEED>;
+    clock-names = "stmmaceth", "mac_clk_rx",
+                 "mac_clk_tx", "clk_mac_ref",
+                 "clk_mac_refout", "aclk_mac",
+                 "pclk_mac", "clk_mac_speed";
+    assigned-clocks = <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC>;
+    assigned-clock-parents = <&cru SCLK_GMAC_RMII_SPEED>, <&gmac_clkin>;
    snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};
```

3.2.3 RGMII Clock Output

```
&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <125000000>;
    snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;
    tx_delay = <0x50>;
    rx_delay = <0x3a>;
    status = "okay";
};
```

3.2.4 RGMII Clock Input

```
&gmac {
```

```

        phy-supply = <&vcc_phy>;
+       phy-mode = "rgmii";
+       clock_in_out = "input";
+       assigned-clocks = <&cru SCLK_GMAC>;
+       assigned-clock-parents = <&gmac_clkin>;
snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;
tx_delay = <0x50>;
rx_delay = <0x3a>;
status = "okay";

};

```

3.3 RK3128

3.3.1 RMII Clock Output

```

&gmac {
+       assigned-clocks = <&cru SCLK_MAC_SRC>;
+       assigned-clock-rates = <50000000>;
+       clock_in_out = "output";
+       pinctrl-names = "default";
+       pinctrl-0 = <&rmii_pins>;
phy-supply = <&vcc_phy>;
+       phy-mode = "rmii";
snps,reset-active-low;
snps,reset-delays-us = <0 10000 50000>;
snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
status = "okay";

};

```

3.3.2 RMII Clock Input

```

+&clkin_gmac {
+       clock-frequency = <50000000>;
+};

&gmac {
+       assigned-clocks = <&cru SCLK_MAC>;
+       assigned-clock-parents = <&clkin_gmac>;
+       clock_in_out = "input";
+       pinctrl-names = "default";
+       pinctrl-0 = <&rmii_pins>;
phy-supply = <&vcc_phy>;
+       phy-mode = "rmii";
snps,reset-active-low;
snps,reset-delays-us = <0 10000 50000>;
snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
status = "okay";

};

```

3.3.3 RGMII Clock Input

```
&gmac {
+   assigned-clocks = <&cru SCLK_MAC>;
+   assigned-clock-parents = <&clkin_gmac>;
+   clock_in_out = "input";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmii_pins>;
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rgmii";
+   snps,reset-active-low;
+   snps,reset-delays-us = <0 20000 100000>;
+   snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
+   tx_delay = <0x30>;
+   rx_delay = <0x16>;
+   status = "okay";
};
```

3.4 RK3228

3.4.1 RMII Clock Output

```
&gmac {
+   assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+   assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+   assigned-clock-rates = <0>, <50000000>;
+   clock_in_out = "output";
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rmii";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii_pins>;
+   snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
+   snps,reset-active-low;
+   snps,reset-delays-us = <0 20000 100000>;
+   status = "okay";
};
```

3.4.2 RMII Clock Input

```
+&ext_gmac: external-gmac-clock {
+   clock-frequency = <50000000>;
+}

&gmac {
+   assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+   assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+   clock_in_out = "input";
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rmii";
+   pinctrl-names = "default";
```

```
+    pinctrl-0 = <&rmii_pins>;
snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
status = "okay";

};
```

3.4.3 RGMII Clock Output

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+    assigned-clock-rates = <0>, <125000000>;
+    clock_in_out = "output";
phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmii_pins>;
snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
tx_delay = <0x30>;
rx_delay = <0x10>;
status = "okay";

};
```

3.4.4 RGMII Clock Input

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+    clock_in_out = "input";
phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmii_pins>;
snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
tx_delay = <0x30>;
rx_delay = <0x10>;
status = "okay";

};
```

3.4.5 Internal EPHY

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC_SRC>;
+    assigned-clock-rates = <50000000>;
+    clock_in_out = "output";
```

```

        phy-supply = <&vcc_phy>;
+       phy-mode = "rmii";
        phy-handle = <&phy>;
        status = "okay";

        mdio {
                compatible = "snps,dwmac-mdio";
                #address-cells = <1>;
                #size-cells = <0>;

                phy: ethernet-phy@0 {
                        compatible = "ethernet-phy-id1234.d400", "ethernet-phy-
ieee802.3-c22";

                        reg = <0>;
                        clocks = <&cru SCLK_MAC_PHY>;
                        resets = <&cru SRST_MACPHY>;
                        phy-is-integrated;

                };
        };
};

```

3.5 RK3288

3.5.1 RMII Clock Output

```

&gmac {
        phy-supply = <&vcc_phy>;
+       phy-mode = "rmii";
+       clock_in_out = "output";
+       assigned-clocks = <&cru SCLK_MAC>;
+       assigned-clock-rates = <50000000>;
        snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;
        snps,reset-active-low;
        snps,reset-delays-us = <0 20000 1000000>;
+       pinctrl-names = "default";
+       pinctrl-0 = <&rmii_pins>;
        status = "okay";
};

```

3.5.2 RMII Clock Input

```

+&ext_gmac: external-gmac-clock {
+       clock-frequency = <50000000>;
+}

&gmac {
        phy-supply = <&vcc_phy>;
+       phy-mode = "rmii";
+       clock_in_out = "input";
+       assigned-clocks = <&cru SCLK_MAC>;
+       assigned-clock-parents = <&ext_gmac>;
        snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;

```



```

        snps,reset-active-low;
        snps,reset-delays-us = <0 20000 1000000>;
+       pinctrl-names = "default";
+       pinctrl-0 = <&rmii_pins>;
        status = "okay";
};

```

3.5.3 RGMII Clock Input

```

&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "input";
    snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 1000000>;
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmii_pins>;
    tx_delay = <0x30>;
    rx_delay = <0x10>;
    status = "okay";
};

```

3.6 RK3328

3.6.1 RMII Clock Output

```

&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC2IO>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii1_pins>;
    status = "okay";
};

```

3.6.2 RMII Clock Input

```

+&clkkin_gmac {
+    clock-frequency = <50000000>;
+};

```

```

&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC2IO>, <&cru SCLK_MAC2IO_EXT>;
+    assigned-clock-parents = <&gmac_clkin>, <&gmac_clkin>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_m1_pins>;
    status = "okay";
};

```

3.6.3 RGMII Clock Input

```

&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC2IO>, <&cru SCLK_MAC2IO_EXT>;
+    assigned-clock-parents = <&gmac_clkin>, <&gmac_clkin>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmii_m1_pins>;
    tx_delay = <0x26>;
    rx_delay = <0x11>;
    status = "okay";
};

```

3.6.4 Internal EPHY

```

&gmac2phy {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC2PHY_SRC>;
+    assigned-clock-rate = <50000000>;
+    assigned-clocks = <&cru SCLK_MAC2PHY>;
+    assigned-clock-parents = <&cru SCLK_MAC2PHY_SRC>;
    status = "okay";
};

```

3.7 RK3368

3.7.1 RMII Clock Output

```

&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "ok";
};

```

3.7.2 RMII Clock Input

```

+&ext_gmac {
+    clock-frequency = <50000000>;
+}

&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "ok";
};

```

3.7.3 RGMII Clock Input

```

&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};

```

3.8 RK3399

3.8.1 RMII Clock Output

```
&gmac {
+   assigned-clocks = <&cru SCLK_MAC>;
+   assigned-clock-rates = <50000000>;
+   clock_in_out = "output";
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rmii";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii_pins>;
+   snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
+   snps,reset-active-low;
+   snps,reset-delays-us = <0 20000 100000>;
+   status = "okay";
};
```

3.8.2 RMII Clock Input

```
+&clkin_gmac {
+   clock-frequency = <50000000>;
+};

&gmac {
+   assigned-clocks = <&cru SCLK_RMII_SRC>;
+   assigned-clock-parents = <&clkin_gmac>;
+   clock_in_out = "input";
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rmii";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii_pins>;
+   snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
+   snps,reset-active-low;
+   snps,reset-delays-us = <0 20000 100000>;
+   status = "okay";
};
```

3.8.3 RGMII Clock Input

```
&gmac {
+   assigned-clocks = <&cru SCLK_RMII_SRC>;
+   assigned-clock-parents = <&clkin_gmac>;
+   clock_in_out = "input";
+   phy-supply = <&vcc_phy>;
+   phy-mode = "rgmii";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmii_pins>;
+   snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
+   snps,reset-active-low;
+   snps,reset-delays-us = <0 20000 100000>;
+   tx_delay = <0x28>;
+   rx_delay = <0x11>;
};
```

```
        status = "okay";
    };
```

3.9 RK3528

3.9.1 GMAC1 RMII Clock 50M Output, PLL 25M Output

```
&gmac1 {
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PC3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 10000 50000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_clk
        &eth_pins>;

    phy-handle = <&rmii1_phy>;
    status = "okay";
};

&mdio1 {
    rmii1_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};
```

3.9.2 GMAC1 RMII Clock 50M Input, PLL 25M Output

```
&gmac1 {
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 10000 50000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_clk
        &eth_pins>;

    phy-handle = <&rmii1_phy>;
```

```

        status = "okay";
    };

    &mdio1 {
        rmiil_phy: phy@1 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x1>;
            clocks = <&cru CLK_GMAC1_VPU_25M>;
        };
    };
};

```

3.9.3 GMAC1 RGMII PLL output 25M for PHY, PLL output 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_rgmii_clk
        &rgmii_rgmii_bus
        &eth_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};

```

3.9.4 GMAC1 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

```

```

snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

tx_delay = <0x30>;
/* rx_delay = <0x3f>; */

pinctrl-names = "default";
pinctrl-0 = <&rgmii_miim
            &rgmii_tx_bus2
            &rgmii_rx_bus2
            &rgmii_rgmii_clk
            &rgmii_rgmii_bus>;

phy-handle = <&rgmii_phy>;
status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.9.5 GMAC1 RGMII PLL output 25M for PHY, RGMII_CLK input 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
                &rgmii_tx_bus2
                &rgmii_rx_bus2
                &rgmii_rgmii_clk
                &rgmii_rgmii_bus
                &rgmii_clk
                &eth_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

```

```

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};

```

3.9.6 GMAC1 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_rgmii_clk
        &rgmii_rgmii_bus
        &rgmii_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.9.7 GMAC0 & FEPHY

GMAC0 与内部 FEPHY 相连，是固定的 RMII，没有模式可以配置；但可以根据硬件原理图配置 PHY led 功能，有 3 个功能 IO 可配置，配置对应 IO 的 iomux 即可，默认配置如下：

```

&rmii0_phy {
    pinctrl-names = "default";
    pinctrl-0 = <&feephy0_led_link &feephy0_led_spd>;
};

```


3.10 RK3562

3.10.1 RMI Clock Output

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PB1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_clk>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

- gmac0m1:

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_clk>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};
```

```
&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

- gmac1(MAC100):

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rmii_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio1 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

3.10.2 RMII Clock Input, PLL out 25M for PHY

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_clk
        &ethm0_pins>;
```

```

        phy-handle = <&rmii_phy>;
        status = "okay";
    };

    &mdio0 {
        rmii_phy: phy@1 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x1>;
            clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
            assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
            assigned-clock-rates = <25000000>;
        };
    };
};

```

- gmac0m1:

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_clk
        &ethm1_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

- gmac1(MAC100):

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;

```

```

/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

pinctrl-names = "default";
pinctrl-0 = <&rmii_pins
            &ethm1_pins>;

phy-handle = <&rmii_phy>;
status = "okay";
};

&mdio1 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

3.10.3 RGMII PLL output 25M for PHY, PLL output 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
                &rgmiim0_tx_bus2
                &rgmiim0_rx_bus2
                &rgmiim0_rgmii_clk
                &rgmiim0_rgmii_bus
                &ethm0_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
    };
};

```

```

        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;

    };

};

```

- gmac0m1

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_rgmii_clk
        &rgmiim1_rgmii_bus
        &ethm1_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

3.10.4 RGMII PLL output 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

```

```

tx_delay = <0x3f>;
/* rx_delay = <0x3f>; */

pinctrl-names = "default";
pinctrl-0 = <&rgmiim0_miim
            &rgmiim0_tx_bus2
            &rgmiim0_rx_bus2
            &rgmiim0_rgmii_clk
            &rgmiim0_rgmii_bus
            &rgmiim0_clk
            &ethm0_pins>;

phy-handle = <&rgmii_phy>;
status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

- gmac0m1

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x39>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
            &rgmiim1_tx_bus2
            &rgmiim1_rx_bus2
            &rgmiim1_rgmii_clk
            &rgmiim1_rgmii_bus
            &rgmiim1_clk
            &ethm1_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {

```

```

compatible = "ethernet-phy-ieee802.3-c22";
reg = <0x1>;
clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
assigned-clock-rates = <25000000>;
};
};

```

3.10.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_rgmii_clk
        &rgmiim1_rgmii_bus>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

- gmac0m1

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

```

```

tx_delay = <0x3f>;
/* rx_delay = <0x3f>; */

pinctrl-names = "default";
pinctrl-0 = <&rgmiim1_miim
            &rgmiim1_tx_bus2
            &rgmiim1_rx_bus2
            &rgmiim1_rgmii_clk
            &rgmiim1_rgmii_bus>;

phy-handle = <&rgmii_phy>;
status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.10.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
                &rgmiim0_tx_bus2
                &rgmiim0_rx_bus2
                &rgmiim0_rgmii_clk
                &rgmiim0_rgmii_bus
                &rgmiim0_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```



```
};
```

- gmac0m1

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x39>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_rgmii_clk
        &rgmiim1_rgmii_bus
        &rgmiim1_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

3.11 RK3568

3.11.1 RMII Clock Output

- gmac0

```
&gmac0 {
+     phy-mode = "rmii";
+     clock_in_out = "output";
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+     assigned-clock-parents = <&cru SCLK_GMAC0_RMII_SPEED>;
+     assigned-clock-rates = <0>, <50000000>;

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
    pinctrl-names = "default";
```

```

        pinctrl-0 = <&gmac0_miim &gmac0_clk inout &gmac0_rx_bus2 &gmac0_tx_bus2>;

        phy-handle = <&rmii_phy0>;
        status = "okay";
    };

    &mdio0 {
        rmii_phy0: phy@0 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x0>;
        };
    };
};

```

- gmac1m0:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>;
+   assigned-clock-rates = <0>, <50000000>;

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m0_miim &gmac1m0_clk inout &gmac1m0_rx_bus2
&gmac1m0_tx_bus2>;

    phy-handle = <&rmii_phy1>;
    status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>;
+   assigned-clock-rates = <0>, <50000000>;

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";

```

```

+       pinctrl-0 = <&gmac1m1_miim &gmac1m1_clkinout &gmac1m1_rx_bus2
&gmac1m1_tx_bus2>;

       phy-handle = <&rmii_phy1>;
       status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.11.2 RMII Clock Input

- gmac0

```

+&gmac0_clkin{
+       clock-frequency = <50000000>;
+};

&gmac0 {
+       phy-mode = "rmii";
+       clock_in_out = "input";

       snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
       snps,reset-active-low;
       snps,reset-delays-us = <0 20000 100000>;

+       assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+       assigned-clock-parents = <&cru SCLK_GMAC0_RMII_SPEED>, <&gmac0_clkin>;

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac0_miim &gmac0_clkinout &gmac0_rx_bus2
&gmac0_tx_bus2>;

       phy-handle = <&rmii_phy0>;
       status = "okay";
};

&mdio0 {
    rmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0:

```

+&gmac1_clkin{
+       clock-frequency = <50000000>;
+};

&gmac1 {

```

```

+     phy-mode = "rmii";
+     clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>, <&gmac1_clkin>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m0_miim &gmac1m0_clkout &gmac1m0_rx_bus2
&gmac1m0_tx_bus2>;

    phy-handle = <&rmii_phy1>;
    status = "okay";
};
&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1:

```

+&gmac1_clkin{
+     clock-frequency = <50000000>;
+};

&gmac1 {
+     phy-mode = "rmii";
+     clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>, <&gmac1_clkin>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m1_miim &gmac1m1_clkout &gmac1m1_rx_bus2
&gmac1m1_tx_bus2>;

    phy-handle = <&rmii_phy1>;
    status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.11.3 RGMII PLL output 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```
&gmac0 {
+     phy-mode = "rgmii";
+     clock_in_out = "output";
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>, <&cru
CLK_MAC0_OUT>;
+     assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>;
+     assigned-clock-rates = <0>, <125000000>, <25000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac0_miim
+                 &gmac0_tx_bus2
+                 &gmac0_rx_bus2
+                 &gmac0_rgmii_clk
+                 &gmac0_rgmii_bus
+                 &eth0_pins>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC0_OUT>;
    };
};
```

- gmac1m0

```
&gmac1 {
+     phy-mode = "rgmii";
+     clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru
CLK_MAC1_OUT>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+     assigned-clock-rates = <0>, <125000000>, <25000000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m0_miim
+                 &gmac1m0_tx_bus2
+                 &gmac1m0_rx_bus2
```

```

+           &gmac1m0_rgmii_clk
+           &gmac1m0_rgmii_bus
+           &eth1m0_pins>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

- gmac1m1

```

&gmac1 {
+    phy-mode = "rgmii";
+    clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru
CLK_MAC1_OUT>;
+    assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+    assigned-clock-rates = <0>, <125000000>, <250000000>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac1m1_miim
+                &gmac1m1_tx_bus2
+                &gmac1m1_rx_bus2
+                &gmac1m1_rgmii_clk
+                &gmac1m1_rgmii_bus
+                &eth1m1_pins>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

3.11.4 RGMII PLL output 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```
&gmac0 {
+     phy-mode = "rgmii";
+     clock_in_out = "input";
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>, <&cru
CLK_MAC0_OUT>;
+     assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>, <&gmac0_clkin>;
+     assigned-clock-rates = <0>, <125000000>, <25000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac0_miim
+                 &gmac0_tx_bus2
+                 &gmac0_rx_bus2
+                 &gmac0_rgmii_clk
+                 &gmac0_rgmii_bus
+                 &eth0_pins
+                 &gmac0_clkinout>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC0_OUT>;
    };
};
```

- gmac1m0

```
&gmac1 {
+     phy-mode = "rgmii";
+     clock_in_out = "input";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru
CLK_MAC1_OUT>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;
+     assigned-clock-rates = <0>, <125000000>, <25000000>;
```

```

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac1m0_miim
+               &gmac1m0_tx_bus2
+               &gmac1m0_rx_bus2
+               &gmac1m0_rgmii_clk
+               &gmac1m0_rgmii_bus
+               &eth1m0_pins
+               &gmac1m0_clkout>;

       tx_delay = <0x4f>;
       rx_delay = <0x26>;

       phy-handle = <&rgmii_phy1>;
       status = "okay";
};

&mdio0 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC0_OUT>;
    };
};

```

- gmac1m1

```

&gmac1 {
+       phy-mode = "rgmii";
+       clock_in_out = "input";

       snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
       snps,reset-active-low;
       /* Reset time is 20ms, 100ms for rtl8211f */
       snps,reset-delays-us = <0 20000 100000>;

+       assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru
CLK_MAC1_OUT>;
+       assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkout>;
+       assigned-clock-rates = <0>, <125000000>, <250000000>;

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac1m1_miim
+               &gmac1m1_tx_bus2
+               &gmac1m1_rx_bus2
+               &gmac1m1_rgmii_clk
+               &gmac1m1_rgmii_bus
+               &eth1m1_pins
+               &gmac1m1_clkout>;

       tx_delay = <0x4f>;
       rx_delay = <0x26>;

       phy-handle = <&rgmii_phy1>;
       status = "okay";
};

&mdio1 {

```



```

    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

3.11.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```

&gmac0 {
+    phy-mode = "rgmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+    assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>;
+    assigned-clock-rates = <0>, <125000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac0_miim
+                &gmac0_tx_bus2
+                &gmac0_rx_bus2
+                &gmac0_rgmii_clk
+                &gmac0_rgmii_bus>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0

```

&gmac1 {
+    phy-mode = "rgmii";
+    clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;

```

```

+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+     assigned-clock-rates = <0>, <125000000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m0_miim
+                 &gmac1m0_tx_bus2
+                 &gmac1m0_rx_bus2
+                 &gmac1m0_rgmii_clk
+                 &gmac1m0_rgmii_bus>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1

```

&gmac1 {
+     phy-mode = "rgmii";
+     clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+     assigned-clock-rates = <0>, <125000000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m1_miim
+                 &gmac1m1_tx_bus2
+                 &gmac1m1_rx_bus2
+                 &gmac1m1_rgmii_clk
+                 &gmac1m1_rgmii_bus>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

```
};  
};
```

3.11.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```
&gmac0 {  
+     phy-mode = "rgmii";  
+     clock_in_out = "input";  
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;  
+     assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>, <&gmac0_clkin>;  
+     assigned-clock-rates = <0>, <125000000>;  
  
     snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;  
     snps,reset-active-low;  
     /* Reset time is 20ms, 100ms for rtl8211f */  
     snps,reset-delays-us = <0 20000 100000>;  
  
+     pinctrl-names = "default";  
+     pinctrl-0 = <&gmac0_miim  
+                 &gmac0_tx_bus2  
+                 &gmac0_rx_bus2  
+                 &gmac0_rgmii_clk  
+                 &gmac0_rgmii_bus  
+                 &gmac0_clkinout>;  
  
     tx_delay = <0x3c>;  
     rx_delay = <0x2f>;  
     phy-handle = <&rgmii_phy0>;  
     status = "okay";  
};  
  
&mdio0 {  
     rgmii_phy0: phy@0 {  
         compatible = "ethernet-phy-ieee802.3-c22";  
         reg = <0x0>;  
     };  
};
```

- gmac1m0

```
&gmac1 {  
+     phy-mode = "rgmii";  
+     clock_in_out = "input";  
  
     snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;  
     snps,reset-active-low;  
     /* Reset time is 20ms, 100ms for rtl8211f */  
     snps,reset-delays-us = <0 20000 100000>;  
  
+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;  
+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;  
+     assigned-clock-rates = <0>, <125000000>;
```

```

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac1m0_miim
+               &gmac1m0_tx_bus2
+               &gmac1m0_rx_bus2
+               &gmac1m0_rgmii_clk
+               &gmac1m0_rgmii_bus
+               &gmac1m0_clkkinout>;

       tx_delay = <0x4f>;
       rx_delay = <0x26>;

       phy-handle = <&rgmii_phy1>;
       status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1

```

&gmac1 {
+       phy-mode = "rgmii";
+       clock_in_out = "input";

       snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
       snps,reset-active-low;
       /* Reset time is 20ms, 100ms for rtl8211f */
       snps,reset-delays-us = <0 20000 100000>;

+       assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+       assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkkin>;
+       assigned-clock-rates = <0>, <125000000>;

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac1m1_miim
+               &gmac1m1_tx_bus2
+               &gmac1m1_rx_bus2
+               &gmac1m1_rgmii_clk
+               &gmac1m1_rgmii_bus
+               &gmac1m1_clkkinout>;

       tx_delay = <0x4f>;
       rx_delay = <0x26>;

       phy-handle = <&rgmii_phy1>;
       status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

```
};
```

3.11.7 SGMII

DTS 除了配置 gmac 和 mac phy 节点外，还需要配置 xpcs 和 combophy 节点。

- combophy

其中属性 `rockchip,sgmii-mac-sel` 表示使用的是哪个 gmac:

```
&combphy1_usq {
+   rockchip,sgmii-mac-sel = <0>; /* Use gmac0 for sgmii */
   status = "okay";
};
```

- xpcs

```
&xpcs {
   status = "okay";
};
```

- gmac0

```
&gmac0 {
   phy-mode = "sgmii";

   rockchip,pipegrf = <&pipegrf>;
   rockchip,xpcs = <&xpcs>;

   snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   snps,reset-delays-us = <0 20000 100000>;

   assigned-clocks = <&cru SCLK_GMAC0_RX_TX>;
   assigned-clock-parents = <&gmac0_xpcscclk>;

   pinctrl-names = "default";
   pinctrl-0 = <&gmac0_miim>;

   power-domains = <&power RK3568_PD_PIPE>;
   phys = <&combphy1_usq PHY_TYPE_SGMII>;
   phy-handle = <&sgmii_phy>;
   status = "okay";
};

&mdio0 {
   sgmii_phy: phy@1 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x1>;
   };
};
```

- gmac1

```

&gmac1 {
    phy-mode = "sgmii";

    rockchip,pipegrf = <&pipegrf>;
    rockchip,xpcs = <&xpcs>;

    snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>;
    assigned-clock-parents = <&gmac1_xpcscclk>;

    pinctrl-names = "default";
    pinctrl-0 = <&gmac1_miim>;

    power-domains = <&power RK3568_PD_PIPE>;
    phys = <&combphy1_usq PHY_TYPE_SGMII>;
    phy-handle = <&sgmii_phy>;
    status = "okay";
};

&mdio1 {
    sgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.11.8 QSGMII

同 SGMII 类似，DTS 除了配置 gmac 和 mac phy 节点外，还需要配置 xpcs 和 combophy 节点。

- combophy

```

&combphy2_psq {
    status = "okay";
};

```

- xpcs

```

&xpcs {
    status = "okay";
};

```

```

&gmac0 {
    phy-supply = <&pcie20_3v3>;
    phy-mode = "qsgmii";
    rockchip,xpcs = <&xpcs>;

    snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

```

```

assigned-clocks = <&cru SCLK_GMAC0_RX_TX>;
assigned-clock-parents = <&gmac0_xpcscclk>;

pinctrl-names = "default";
pinctrl-0 = <&gmac0_miim>;

power-domains = <&power RK3568_PD_PIPE>;
phys = <&combphy2_psq PHY_TYPE_QSGMII>;
phy-handle = <&qsgmii_phy0>;

status = "okay";
};

&gmac1 {
    phy-supply = <&pcie20_3v3>;
    phy-mode = "qsgmii";

    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>;
    assigned-clock-parents = <&gmac1_xpcscclk>;

    power-domains = <&power RK3568_PD_PIPE>;
    phy-handle = <&qsgmii_phy1>;

    status = "okay";
};

&mdio0 {
    qsgmii_phy0: phy@0 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
    qsgmii_phy1: phy@1 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
    qsgmii_phy2: phy@2 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x2>;
    };
    qsgmii_phy3: phy@3 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x3>;
    };
};

```

3.12 RK3588/RK3576

3.12.1 RMII Clock Output

- gmac0

```

&gmac0 {
+     phy-mode = "rmii";
+     clock_in_out = "output";

```

```

snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_clkout>;

phy-handle = <&rmii_phy0>;
status = "okay";
};

&mdio0 {
    rmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

- gmac1:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "output";

snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_clkout>;

phy-handle = <&rmii_phy1>;
status = "okay";
};

&mdio1 {
    rmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.12.2 RMII Clock Input

- gmac0


```

&gmac0 {
+   phy-mode = "rmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_clkout>;

   phy-handle = <&rmii_phy0>;
   status = "okay";
};

&mdio0 {
   rmii_phy0: phy@1 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x1>;
   };
};

```

- gmac1:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_clkout>;

   phy-handle = <&rmii_phy1>;
   status = "okay";
};

&mdio1 {
   rmii_phy1: phy@1 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x1>;
   };
};

```

3.12.3 RGMII PLL output 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+       &gmac0_tx_bus2
+       &gmac0_rx_bus2
+       &gmac0_rgmii_clk
+       &gmac0_rgmii_bus
+       &eth0_pins>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
+       clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};
```

- gmac1

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
+   clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+       &gmac1_tx_bus2
+       &gmac1_rx_bus2
+       &gmac1_rgmii_clk
+       &gmac1_rgmii_bus
+       &eth1_pins>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */
```

```

        phy-handle = <&rgmii_phy1>;
        status = "okay";
    };

    &mdio1 {
        rgmii_phy1: phy@1 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x1>;
+           clocks = <&cru REFCLK025M_ETH0_OUT>;
        };
    };
};

```

3.12.4 RGMII PLL output 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_rgmii_clk
+               &gmac0_rgmii_bus
+               &gmac0_clkinout
+               &eth0_pins>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
+       clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};

```

- gmac1

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */

```

```

+   phy-mode = "rgmii-rxid";
+   clock_in_out = "input";

snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_rgmii_clk
+               &gmac1_rgmii_bus
+               &gmac1_clkinout
+               &eth1_pins>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
+       clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};
};

```

3.12.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
+   clock_in_out = "output";

snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_rgmii_clk
+               &gmac0_rgmii_bus>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

```

```

        phy-handle = <&rgmii_phy0>;
        status = "okay";
    };

    &mdio0 {
        rgmii_phy0: phy@0 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x0>;
        };
    };
};

```

- gmac1

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    + phy-mode = "rgmii-rxid";
    + clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rt18211f */
    snps,reset-delays-us = <0 20000 100000>;

    + pinctrl-names = "default";
    + pinctrl-0 = <&gmac1_miim
    +             &gmac1_tx_bus2
    +             &gmac1_rx_bus2
    +             &gmac1_rgmii_clk
    +             &gmac1_rgmii_bus>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.12.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    + phy-mode = "rgmii-rxid";
    + clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;

```

```

snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_rgmii_clk
+               &gmac0_rgmii_bus
+               &gmac0_clkinout>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy0>;
status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
+   clock_in_out = "input";

snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_rgmii_clk
+               &gmac1_rgmii_bus
+               &gmac1_clkinout>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

```
};  
};
```

3.13 RV1108

3.13.1 RMII Clock Input

```
+gmac_clkin: gmac_clkin {  
+    compatible = "fixed-clock";  
+    clock-output-names = "gmac_clkin";  
+    clock-frequency = <50000000>;  
+    #clock-cells = <0>;  
+};  
  
&gmac {  
+    phy-mode = "rmii";  
+    clock_in_out = "input";  
+    assigned-clocks = <&cru SCLK_MAC>;  
+    assigned-clock-parents = <&gmac_clkin>;  
+    snps,reset-gpio = <&gpio3 12 0>;  
+    snps,reset-active-low;  
+    snps,reset-delays-us = <0 20000 100000>;  
+    pinctrl-names = "default";  
+    pinctrl-0 = <&rmii_pins>;  
+    status = "ok";  
};
```

3.13.2 RMII Clock Output

```
&gmac {  
+    phy-mode = "rmii";  
+    clock_in_out = "output";  
+    assigned-clocks = <&cru SCLK_MAC>;  
+    assigned-clock-rates = <50000000>;  
+    snps,reset-gpio = <&gpio3 12 0>;  
+    snps,reset-active-low;  
+    snps,reset-delays-us = <0 20000 100000>;  
+    pinctrl-names = "default";  
+    pinctrl-0 = <&rmii_pins>;  
+    status = "ok";  
};
```

3.14 RV1126

3.14.1 RGMII PLL output 25M for PHY, PLL output 125M for TX_CLK

- gmac m0

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

- gmac m1

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkm1_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

```



```

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

3.14.2 RGMII PLL output 25M for PHY, RGMII Clock input 125M for TX_CLK

- gmac m0

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

    tx_delay = <0x2a>;
    rx_delay = <0x1a>;

    phy-handle = <&phy>;
    status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

- gmac m1

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

```

```

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkml_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

3.14.3 RGMII Crytal 25M for PHY, PLL output 125M for TX_CLK

- gmac m0

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;

```

```
};
};
```

- gmac m1

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkml_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
   phy: phy@0 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x0>;
   };
};
```

3.14.4 RGMII Crtal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac m0

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>, <&cru
CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-parents = <&gmac_clkin_m0>, <&cru CLK_GMAC_RGMII_M0>, <&cru
CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4>;
```

```

        tx_delay = <0x2a>;
        rx_delay = <0x1a>;

        phy-handle = <&phy>;
        status = "okay";
    };

    &mdio {
        phy: phy@0 {
            compatible = "ethernet-phy-ieee802.3-c22";
            reg = <0x0>;
        };
    };
};

```

- gmac m1

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4>;

    tx_delay = <0x2a>;
    rx_delay = <0x1a>;

    phy-handle = <&phy>;
    status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.14.5 RMII Clock Output

- gmac m0

```

&gmac {
+   phy-mode = "rmii";
+   clock_in_out = "output";

```

```

snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 50000 50000>;

+ assigned-clocks = <&cru CLK_GMAC_SRC_M0>, <&cru CLK_GMAC_SRC>, <&cru
CLK_GMAC_TX_RX>;
+ assigned-clock-rates = <0>, <500000000>;
+ assigned-clock-parents = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>,
<&cru RMII_MODE_CLK>;

+ pinctrl-names = "default";
+ pinctrl-0 = <&rmii0_miim &rgmii0_rxer &rmii0_bus2 &rgmii0_mclkinout>;

phy-handle = <&phy>;
status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac m1

```

&gmac {
+ phy-mode = "rmii";
+ clock_in_out = "output";

snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 50000 50000>;

+ assigned-clocks = <&cru CLK_GMAC_SRC_M1>, <&cru CLK_GMAC_SRC>, <&cru
CLK_GMAC_TX_RX>;
+ assigned-clock-rates = <0>, <500000000>;
+ assigned-clock-parents = <&cru CLK_GMAC_RGMII_M1>, <&cru CLK_GMAC_SRC_M1>,
<&cru RMII_MODE_CLK>;

+ pinctrl-names = "default";
+ pinctrl-0 = <&rmii1_miim &rgmii1_rxer &rmii10_bus2 &rgmii1_mclkinout>;

phy-handle = <&phy>;
status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.14.6 RMII Clock Input

- gmac m0

```
+&gmac_clkin_m0 {
+  clock-frequency = <50000000>;
+};

&gmac {
+  phy-mode = "rmii";
+  clock_in_out = "input";

  snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
  snps,reset-active-low;
  snps,reset-delays-us = <0 50000 50000>;

+  assigned-clocks = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>, <&cru
CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+  assigned-clock-rates = <0>, <0>, <50000000>;
+  assigned-clock-parents = <&gmac_clkin_m0>, <&cru CLK_GMAC_RGMII_M0>, <&cru
CLK_GMAC_SRC_M0>, <&cru RMII_MODE_CLK>;

+  pinctrl-names = "default";
+  pinctrl-0 = <&rmii0_miim &rgmii0_rxer &rmii0_bus2
&rgmii0_mclkinout_level0>;

  phy-handle = <&phy>;
  status = "okay";
};

&mdio {
  phy: phy@0 {
    compatible = "ethernet-phy-ieee802.3-c22";
    reg = <0x0>;
  };
};
```

- gmac m1

```
+&gmac_clkin_m1 {
+  clock-frequency = <50000000>;
+};

&gmac {
+  phy-mode = "rmii";
+  clock_in_out = "input";

  snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
  snps,reset-active-low;
  snps,reset-delays-us = <0 50000 50000>;

+  assigned-clocks = <&cru CLK_GMAC_RGMII_M1>, <&cru CLK_GMAC_SRC_M1>, <&cru
CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+  assigned-clock-rates = <0>, <0>, <50000000>;
+  assigned-clock-parents = <&gmac_clkin_m1>, <&cru CLK_GMAC_RGMII_M1>, <&cru
CLK_GMAC_SRC_M1>, <&cru RMII_MODE_CLK>;

+  pinctrl-names = "default";
```

```
+   pinctrl-0 = <&rmii1_miim &rgmii1_rxer &rmii1_bus2
&rgmii1_mclkinout_level0>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```